🏛 ENTERPRISE EDITION

System Guide • October 2026

Athena Library Management System

Complete Operational Handbook & Day-to-Day Staff Workflow Manual

01 System Overview & Cloud Architecture

Athena LMS is a full-stack, enterprise-grade library management system. It is designed to automate physical inventory cataloging, streamline book circulation, eliminate overdue tracking paperwork, and deliver 24/7 uptime.

Responsive UI

Optimized for desktops, circulation desk monitors, tablets, and smartphones.

Java Server

Multi-threaded native Java backend engine with high-concurrency connection pooling.

Supabase Cloud

PostgreSQL cloud storage with live cross-device synchronization and foreign key integrity.

SQLite Fallback

Embedded database failover guaranteeing continuous operation even if cloud pauses.

02 User Roles & Access Hierarchy

ROLE	CREDENTIALS / ACCESS	PERMISSIONS & OPERATIONAL SCOPE
Administrator	`sa@gmail.com` `shamilc12`	Full system governance: Catalog new books, issue & return books, manage users, configure fine engines, view circulation audit logs and analytics.
Librarian / Staff	Assigned by Administrator	Circulation desk operations: Issue loans, process returns, calculate late fees, search shelf coordinates.
Student Patron	Self-registered via portal	Catalog discovery: Search books by title/author/ISBN, view physical availability, inspect personal active loans and due dates.

🔑 **Administrator Master Account**

The primary admin account (**sa@gmail.com / shamilc12**) is permanently pre-seeded into the database. You will never be locked out even during server container redeployments.

03 Daily Library Operations (Step-by-Step)

STEP 1 Starting the Workday & Staff Login

- Open your library URL on the circulation desk PC or tablet.
- Click **"Client Login"** or **"Enter System Dashboard"**.
- Enter your staff or administrator credentials and click **"Sign In to Dashboard"**.
- *Security feature:* The login form automatically clears upon loading and locks out third-party browser autofill to prevent unauthorized access.

STEP 2 Cataloging New Books & Shelf Placement

The system separates **Book Titles** from **Physical Copies** (e.g., 1 title can have 5 physical copies with individual barcodes).

- Navigate to **"Book Catalog"** from the menu.
- Click the gold **"+ Add New Book"** button.
- Fill in the metadata: *Title*, *Author*, *ISBN*, *Category*, *Total Copies* (e.g. 3), and *Shelf Coordinate* (e.g. `Rack B-2, Shelf 4`).
- Click **"Save Book"**. The system automatically creates the book title and generates 3 available physical copy records assigned to that shelf coordinate.

STEP 3 Registering Members & Students

- **Self-Service:** Students can open the website, switch to the **"Register"** tab, enter their Name, Email, and a password (minimum 6 characters with both letters & numbers).
- **Staff Verification:** Librarians can navigate to the **"Members"** directory to view all registered patrons, their unique IDs (e.g. `U1`, `U2`), and their active loan counts.

STEP 4 Issuing a Book to a Member (Borrowing Desk)

When a patron brings a physical book to the circulation counter:

- Click **"Issue Book"** on the Dashboard or Catalog screen.
- **Member Identifier:** Type the patron's *Member ID* (e.g. `U1`), *Email* (`student@example.com`), or *Full Name*.
- **Book Identifier:** Type the *Book ID*, *ISBN*, or scan the *Copy Barcode*.
- Click **"Confirm Issue"**.
- **Validation Engine:** The system verifies that the student has not exceeded the **5-book maximum borrowing limit** and that a copy is physically available. It sets an automated **14-day due date** and marks the copy as `ISSUED`.

STEP 5 Processing Returns & Automated Fine Engine

When a patron returns a borrowed book:

- Click **"Return Book"** at the circulation desk.
- Enter the *Book ID*, *Copy Barcode*, or *Transaction ID*.
- Click **"Process Return"**.
- **On-Time Return:** The book copy is immediately restored to `AVAILABLE` status, the transaction is marked `RETURNED`, and celebration confetti displays on screen.
- **Late Return (Fine Engine):** If the book is returned past its 14-day due date, the system calculates late fines at **₹5 per overdue day** and logs an unpaid fine record linked to the student.

STEP 6 Dashboard Metrics & Overdue Tracking

- **KPI Cards:** Real-time counts of *Total Books*, *Active Loans*, *Overdue Books*, and *Total Members*.
- **Circulation Trend Analytics:** Chart.js visualization showing monthly circulation patterns to assist in book acquisition planning.
- **Live Activity Feed:** Real-time audit log recording every loan checkout and return timestamp.

04 Physical Library Setup & Best Practices

HARDWARE / PRACTICE	SETUP DETAILS	BENEFIT TO LIBRARY STAFF
Barcode Scanner	Standard USB or Bluetooth 1D/2D laser barcode scanner plugged into the desk PC.	Scanners act as rapid keyboard entry: scanning a book barcode instantly populates the Issue and Return fields with zero typing errors.

HARDWARE / PRACTICE	SETUP DETAILS	BENEFIT TO LIBRARY STAFF
Shelf Rack Labeling	Print physical stickers for shelves matching your catalog coordinates (e.g. `Rack A, Shelf 1`).	Students and library assistants can locate any title in under 30 seconds by checking the "Shelf Location" column in the catalog.
Mobile Aisle Tablet	iPad, Android tablet, or smartphone connected to Wi-Fi.	Staff can perform physical inventory audits directly in the library aisles without carrying books back to the main desk.

05 Circulation Rules Quick Reference

📋 **Circulation Engine Parameters**

- **Standard Borrowing Period:** 14 Calendar Days
- **Maximum Loans per Patron:** 5 Books simultaneously
- **Late Fine Rate:** ₹5.00 per day past the due date
- **Search Filters:** Real-time instant filtering across Title, Author, Category, and ISBN
- **Data Storage:** Dual PostgreSQL (Supabase) + Local SQLite Fallback